

Software Engineer Onboarding Manual

Synthetic Edition 03 • Fictional company, people, systems, and resources

Welcome to BluePeak Technologies! You are joining an engineering organization that builds workflow automation, customer portals, internal tools, and cloud services. This manual provides a practical introduction to our teams, products, engineering practices, and first-month expectations.

Note: All information in this document is synthetic and intended for demonstrations, testing, training, and document-generation examples.

1. Welcome to BluePeak

Starting a new engineering role can involve many unfamiliar systems and processes. During onboarding, focus on learning how the organization works, establishing your development environment, and building relationships with the people who can help you succeed.

Our Working Principles

- **Customer first:** Understand the problem before choosing a technical solution.
- **Quality by default:** Testing and maintainability are part of development.
- **Small changes:** Prefer focused pull requests that are easy to review.
- **Transparent communication:** Share progress, risks, and blockers early.
- **Continuous improvement:** Improve the codebase and the process around it.

2. Engineering Leadership

Position	Name	Handle	Responsibility
Chief Technology Officer	Elena Rodriguez	@elena-r	Technology strategy
Engineering Manager	Marcus Lee	@marcus-l	Team delivery and people development
Backend Lead	Owen Carter	@owen-c	APIs and service architecture
Frontend Lead	Mina Desai	@mina-d	Web applications and design systems
QA Lead	Grace Wilson	@grace-w	Quality engineering
Cloud Lead	Noah Bennett	@noah-b	Infrastructure and reliability
Data Lead	Lucas Martin	@lucas-m	Analytics and data platforms

Who to Contact

- **Mentor:** General onboarding and day-to-day questions.
- **Manager:** Goals, priorities, role expectations, and career development.
- **Project lead:** Product requirements and project decisions.
- **QA lead:** Testing strategy and release quality.
- **Cloud lead:** Deployment, infrastructure, monitoring, and reliability.

3. Product Areas

Product	Purpose	Area	Lead	Status
BeaconDesk	Customer support and case management	Web	Mina Desai	Active
FlowGrid	Business workflow automation	Platform	Owen Carter	Active

Product	Purpose	Area	Lead	Status
ClearView	Operational analytics and reporting	Data	Lucas Martin	Active
LaunchPad	Internal release management portal	Cloud	Noah Bennett	Planning
Bridge	Partner API integration platform	Backend	Owen Carter	Active
CanvasPro	Shared UI and accessibility system	Frontend	Mina Desai	Maintenance
Watchtower	Application health and alerting	Cloud	Noah Bennett	Active
LedgerLine	Subscription and invoice reconciliation	Backend/Data	Owen Carter	Planning

How Projects Are Assigned

New engineers are assigned according to team capacity, technical interests, product priorities, and onboarding requirements. Your first task is intentionally small so you can learn the team's workflow before taking on larger work.

4. New Engineer Directory

ID	Engineer	Role	Product	Level	Mentor
BP-301	Rhea Kapoor	Backend Engineer	FlowGrid	Senior	Owen Carter
BP-302	Zayn Mitchell	Frontend Engineer	BeaconDesk	Mid-Level	Mina Desai
BP-303	Aarohi Shah	QA Automation Engineer	Bridge	Mid-Level	Grace Wilson
BP-304	Ethan Brooks	Cloud Engineer	Watchtower	Senior	Noah Bennett
BP-305	Ishaan Roy	Data Engineer	ClearView	Junior	Lucas Martin
BP-306	Nora Patel	Full-Stack Engineer	BeaconDesk	Mid-Level	Mina Desai
BP-307	Caleb Foster	Backend Engineer	Bridge	Junior	Owen Carter
BP-308	Mira Joshi	Data Analyst	ClearView	Mid-Level	Lucas Martin
BP-309	Theo Morgan	Platform Engineer	FlowGrid	Senior	Noah Bennett
BP-310	Sia Fernandes	Frontend Engineer	CanvasPro	Junior	Mina Desai
BP-311	Ryan Cooper	QA Engineer	LedgerLine	Mid-Level	Grace Wilson
BP-312	Ira Kulkarni	SRE Engineer	Watchtower	Senior	Noah Bennett

5. Key Resources

- **Engineering Hub:** <https://example.com/bluepeak/engineering>
- **Source Repositories:** <https://example.com/bluepeak/repos>
- **Knowledge Base:** <https://example.com/bluepeak/docs>
- **Issue Tracker:** <https://example.com/bluepeak/work>
- **Design Library:** <https://example.com/bluepeak/design>
- **CI Dashboard:** <https://example.com/bluepeak/builds>
- **Observability:** <https://example.com/bluepeak/observe>
- **People Portal:** <https://example.com/bluepeak/people>

The example.com addresses above are placeholders and should be replaced with real organizational links when adapting this guide.

6. Technology Stack

- **Frontend:** React, TypeScript, CSS, component libraries

- **Backend:** Java, Spring Boot, Python, REST APIs
- **Data:** PostgreSQL, Redis, SQL, scheduled pipelines
- **Testing:** JUnit, Pytest, Playwright, API testing
- **Infrastructure:** Containers, cloud services, infrastructure-as-code
- **Observability:** Centralized logs, metrics, traces, and alerts
- **Version control:** Git with pull-request based development

Development Setup

- Install Git and configure approved authentication.
- Install the supported IDE.
- Install the runtime required by your project.
- Install Docker or the approved container runtime.
- Configure local environment variables.
- Clone the assigned repository.
- Run the project's test suite.
- Start the application locally.
- Verify the health or status endpoint.

7. First Week Schedule

Day	Focus	Activities
Day 1	Orientation	HR, account setup, security briefing, workstation setup, mentor meeting.
Day 2	Product Discovery	Product demo, domain overview, architecture presentation.
Day 3	Repository Tour	Codebase walkthrough, local setup, issue-tracker orientation.
Day 4	Engineering Workflow	Git, pull requests, testing, CI/CD, deployment overview.
Day 5	First Contribution	Small task, review, feedback session, next-week planning.

First Week Checklist

- Complete administrative onboarding.
- Activate multi-factor authentication.
- Join required engineering channels.
- Meet your mentor and project lead.
- Read the engineering handbook.
- Read your product's README.
- Run the project locally.
- Review recent pull requests.
- Complete required security training.
- Open a first draft pull request.

8. Engineering Development Workflow

Work Items

Each task should explain the problem, expected behavior, acceptance criteria, and priority. If requirements are unclear, ask for clarification before implementing assumptions.

Branches and Pull Requests

- Use one short-lived branch per meaningful change.
- Use names such as `feature/customer-search` or `fix/login-timeout`.
- Keep pull requests focused.
- Describe motivation, implementation, and testing.
- Resolve review discussions before merging.

Example:

```
git checkout main
git pull origin main
git checkout -b feature/customer-search
git add .
git commit -m "Add customer search"
git push -u origin feature/customer-search
```

9. Testing Standards

Layer	Purpose	Example
Unit	Validate isolated logic	Service or utility test
Integration	Validate component interactions	API + database test
Contract	Validate API expectations	Request/response contract
End-to-End	Validate user workflows	Login and search flow
Regression	Protect existing behavior	Release regression suite

Before Merge

- Code is formatted and linted.
- Relevant automated tests pass.
- New behavior has appropriate tests.
- The author has reviewed the diff.
- Documentation is updated when needed.
- Required approvals are present.

10. Release and Deployment

BluePeak uses automated pipelines to build, test, and package applications. Changes normally progress from development to staging and then production.

- Do not manually change production resources unless the incident process requires it.
- Check deployment status after significant releases.
- Use feature flags when supported for higher-risk functionality.
- Record rollback or mitigation steps for elevated-risk changes.

Production Incident Basics

- Confirm the alert and affected service.
- Determine whether customers are impacted.
- Notify the responsible on-call team.
- Record observations and actions in the incident channel.
- Prefer safe mitigation over large unreviewed changes.
- Document the resolution and follow-up work.

11. Security Guidelines

- Never commit credentials or access tokens.
- Use approved secret-management tools.
- Use synthetic or authorized data in development.
- Do not share private customer information in public channels.
- Use least-privilege permissions.
- Report suspicious activity immediately.
- Keep dependencies and development tools updated according to team policy.

12. Communication

Daily Stand-Up

Share what you completed, what you are working on, and what is blocking you. Move detailed technical discussions into a follow-up conversation.

Asking for Help

A useful technical question contains enough information for someone else to reproduce or understand the problem.

Problem: BeaconDesk fails to start locally.
Expected: Service starts on port 8080.
Actual: Connection to PostgreSQL is refused.
Tried: Restarted containers and verified the environment file.
Question: Is there a required database initialization command?

Synthetic Team Channels

- #bluepeak-engineering — general engineering updates
- #dev-help — technical troubleshooting
- #beacondesk — product development
- #flowgrid — workflow platform
- #quality-hub — QA and testing
- #cloud-ops — infrastructure and reliability
- #data-lab — analytics and data engineering

13. Mentor Program

Mentor	Role	Experience	Primary Topics
Owen Carter	Backend Lead	8 years	APIs, Java, service architecture
Mina Desai	Frontend Lead	6 years	React, TypeScript, accessibility
Grace Wilson	QA Lead	9 years	Automation, quality, release testing

Mentor	Role	Experience	Primary Topics
Noah Bennett	Cloud Lead	7 years	Cloud, CI/CD, observability
Lucas Martin	Data Lead	6 years	SQL, pipelines, analytics

Mentor Meeting Plan

- Day 1 — introduction and onboarding goals
- Day 3 — project architecture and questions
- Day 5 — first-week review
- Week 2 — first contribution feedback
- Week 4 — progress and development discussion

14. 30-Day Roadmap

Period	Focus	Expected Outcome
Days 1–5	Orientation	Understand team, tools, product, and local setup.
Week 2	Guided Contribution	Complete a small task with close review.
Week 3	Independent Work	Own a small issue from implementation through review.
Week 4	Ownership	Deliver a small improvement and present what you learned.

15. Frequently Asked Questions

What should I do if I am blocked?

Check the project documentation and existing issues. Then ask in the appropriate help channel with the context of what you tried. Tell your mentor or project lead if the blocker affects a deadline.

Can I change the architecture?

Architectural changes should be discussed with the project lead and relevant technical lead before implementation.

Should every change have tests?

Production behavior should have appropriate automated or manual validation. The exact level depends on the risk and type of change.

Where should I record important knowledge?

Prefer shared project documentation so future engineers can benefit from the information.

16. First-Month Review

- I understand the purpose of my assigned product.
- I can navigate the repository.
- I can run the project locally.
- I understand the branch and pull-request workflow.
- I have participated in code review.
- I have written or updated tests.
- I understand the basic deployment process.
- I know the appropriate escalation contacts.

- I have completed at least one reviewed contribution.
- I have discussed my next development goals with my manager.

17. Welcome to BluePeak Technologies

We are excited to have you here. Take time to learn the systems around you, communicate clearly, ask questions early, and leave every part of the engineering environment a little better than you found it.

BluePeak Technologies Engineering • Synthetic Edition 03 • 2026